

JavaScript Roadmap for MERN (Beginner to Advanced)

Phase 1: JavaScript Fundamentals (Week 1-2)

1. Basics

Learn:

- What is JavaScript?
- How JS works in the browser
- Variables (let, const, var)
- Data types
- Operators
- Type conversion

Practice

Build:

- Age Calculator
 - BMI Calculator
 - Percentage Calculator
 - Temperature Converter
-

Phase 2: Conditions & Loops (Week 2)

Learn:

- if
- else
- switch
- Ternary Operator
- for
- while
- do while
- break
- continue

Practice

Build:

- Number Guessing Game
 - ATM Machine Simulation
 - Student Grade Calculator
 - Multiplication Table Generator
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Phase 3: Functions (Week 3)

Learn:

- Function Declaration
- Function Expression
- Arrow Functions
- Parameters
- Return

- Default Parameters
- Rest Parameters
- Spread Operator

Practice

Build:

- Calculator using functions
 - Password Generator
 - OTP Generator
 - Dice Roll Simulator
-

Phase 4: Arrays (Week 4)

Learn:

- Create arrays
- Access elements
- Loop through arrays
- Array methods:
 - push()
 - pop()
 - shift()
 - unshift()
 - slice()
 - splice()
 - concat()
 - includes()
 - indexOf()

Higher-order methods

- map()
- filter()
- reduce()
- find()
- some()
- every()
- sort()

Practice

Build:

- Student Marks Manager
 - Shopping Cart
 - Expense Tracker
 - To-Do List
-

Phase 5: Objects (Week 5)

Learn:

- Objects

- Nested Objects
- Object Methods
- this
- Destructuring
- Spread Operator
- Object Keys
- Values
- Entries

Practice

Build:

- Student Database
 - Employee Management System
 - Product Inventory
-

Phase 6: Strings (Week 5)

Learn:

- String methods
- Template literals
- Regular Expressions (basics)

Practice

Build:

- Password Strength Checker
 - Email Validator
 - Username Validator
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Intermediate JavaScript

Phase 7: DOM Manipulation (Week 6)

This is one of the most important topics.

Learn:

- getElementById
- querySelector
- querySelectorAll
- Create elements
- Remove elements
- Modify elements
- CSS through JavaScript

Practice

Build:

- Notes App
- Calculator
- Stopwatch
- Digital Clock
- Image Slider

Phase 8: Events

Learn:

- Click
- Double Click
- Mouse Events
- Keyboard Events
- Form Events
- Event Bubbling
- Event Delegation

Practice

Build:

- Form Validation
 - Color Changer
 - Keyboard Typing Test
-

Phase 9: ES6+ Features

Learn:

- Arrow Functions
- Template Literals
- Destructuring
- Spread
- Rest
- Modules
- Classes
- Promises
- Async/Await

Practice

Convert old JavaScript code into ES6 syntax.

Phase 10: Asynchronous JavaScript

Learn:

- Callbacks
- Callback Hell
- Promises
- Fetch API
- Async/Await
- JSON
- Error Handling

Practice

Build:

- Weather App
- Movie Search App

- GitHub User Finder
 - Currency Converter
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Advanced JavaScript

Phase 11: Advanced Concepts

Learn:

- Closures
- Lexical Scope
- Hoisting
- Execution Context
- Call Stack
- Event Loop
- Prototype
- Prototypal Inheritance
- Classes
- this
- call()
- apply()
- bind()

Practice

Implement small examples for each concept.

Phase 12: Browser Storage

Learn:

- Cookies (basics)
- Local Storage
- Session Storage

Practice

Build:

- Dark Mode Toggle
 - Notes App with saved data
 - Login System (front-end only)
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Phase 13: Error Handling

Learn:

- try
- catch
- finally
- throw

Practice

Add proper validation and error messages to your projects.

JavaScript for MERN

Now learn JavaScript specifically for backend development.

Node.js Basics

Learn:

- Modules
- require
- import
- File System
- Path
- HTTP Module
- NPM
- Package.json

Practice

- Read/write files
 - Build a simple HTTP server
 - Create a CLI application
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Express.js

Learn:

- Express setup
- Routes
- Middleware
- Request & Response
- REST APIs
- CRUD Operations

Practice

Build:

- Notes API
 - Student Management API
 - Product API
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MongoDB

Learn:

- Collections
- Documents
- CRUD
- Queries
- Aggregation (basics)

Practice

Store and retrieve data for your Express projects.

React Preparation

Before React, make sure you're comfortable with:

- ES6+

- Array methods (map, filter, reduce)
- Destructuring
- Spread operator
- Modules
- Promises
- Async/Await
- Fetch API
- DOM and events